Proprietary + Confidential

06

Automation techniques

Proprietary + Confidential

Automation techniques

01	Automation patterns and options for pipelines
02	Cloud Scheduler and Workflows
03	Cloud Composer
04	Cloud Run functions
05	Eventarc

Google Cloud

In this module, first, you review the automation patterns and options available for pipelines.

Second, you explore Cloud Scheduler and workflows.

Then, you review the functionality and use cases for Cloud Composer.

Next, you review the capabilities of Cloud Run functions.

Finally, you look at the functionality and automation use cases for Eventarc.

Proprietary + Confidential

Automation techniques

01 Automation patterns and options for pipelines

02 Cloud Scheduler and Workflows

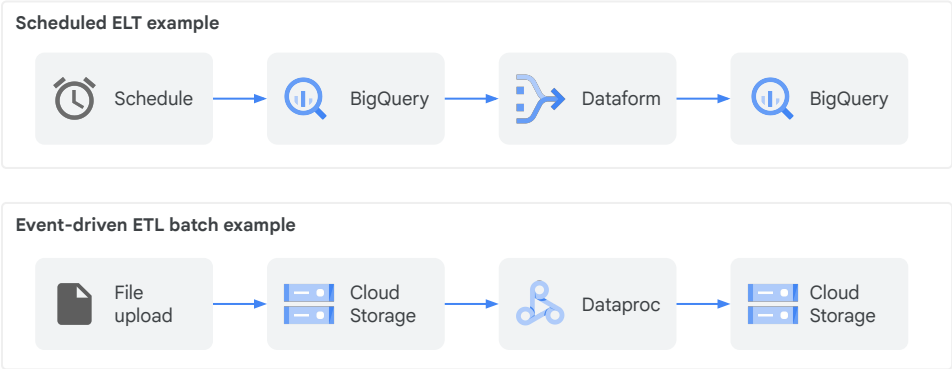
03 Cloud Composer

04 Cloud Run functions

05 Eventarc

Proprietary + Confidential

ELT and ETL workloads can be automated to run on a recurring basis

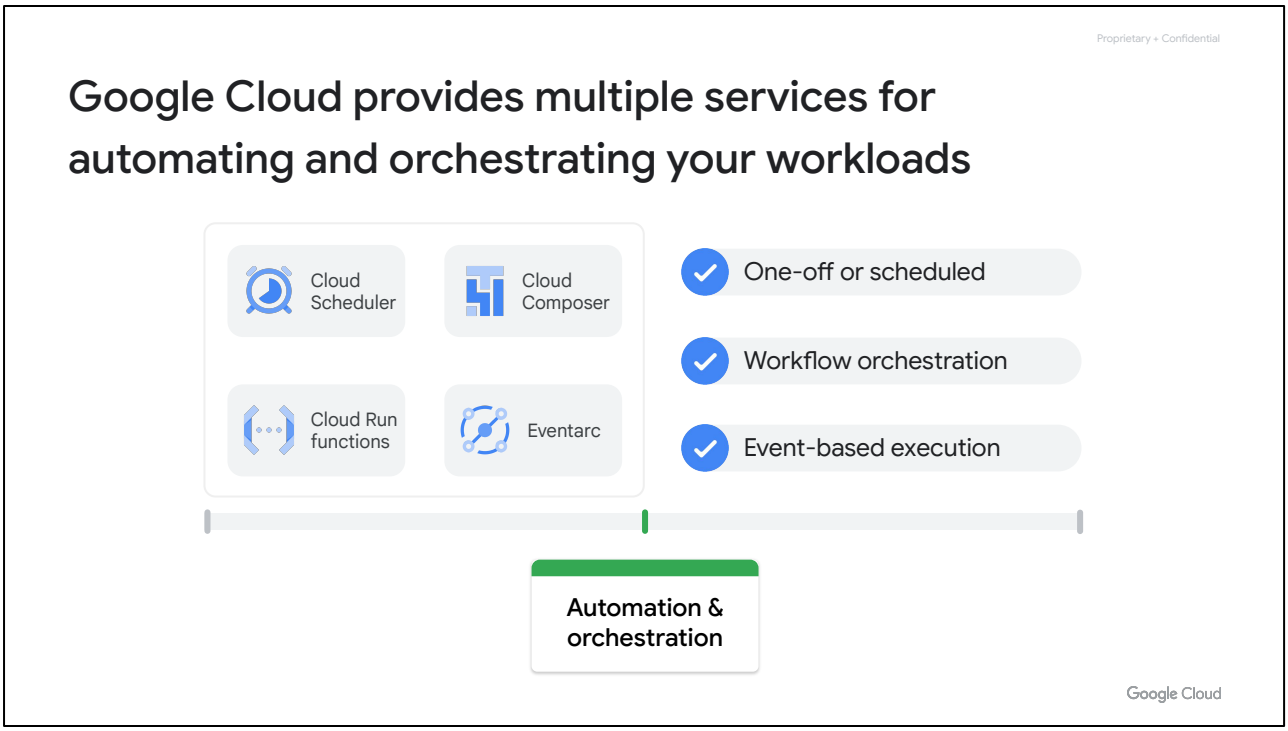


Google Cloud

On Google Cloud, ELT and ETL workloads can be automated for recurring execution.

For example, in a scheduled ELT, a defined schedule triggers data extraction from BigQuery, transformation via Dataform, and loading back into BigQuery.

Meanwhile, in the example shown for an event-driven ETL, a file upload to Cloud Storage initiates a batch process using Dataproc, culminating in data landing in Cloud Storage.



Google Cloud offers a suite of services to automate and orchestrate your workloads.

For scheduled tasks or one-off jobs, you can leverage Cloud Scheduler and Cloud Composer.

If your workflows require orchestration, Cloud Composer is the ideal choice.

To trigger actions based on events, consider using Cloud Run functions or Eventarc.

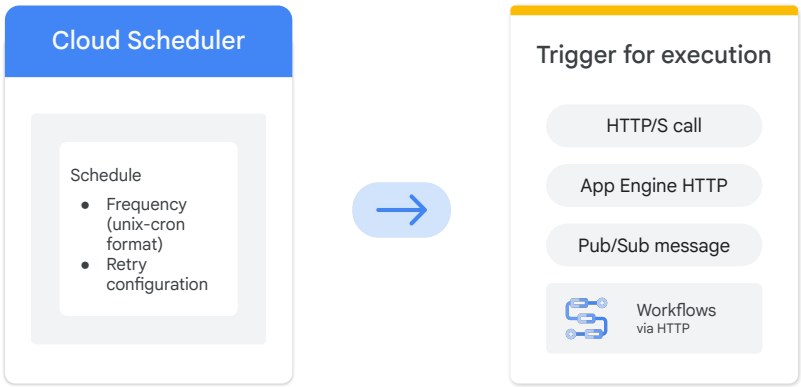
Proprietary + Confidential

Automation techniques

01	Automation patterns and options for pipelines
02	Cloud Scheduler and Workflows
03	Cloud Composer
04	Cloud Run functions
05	Eventarc

Proprietary + Confidential

Cloud Scheduler invokes your workloads at recurring intervals



Google Cloud

Cloud Scheduler empowers you to automate tasks by invoking your workloads at specified, recurring intervals.

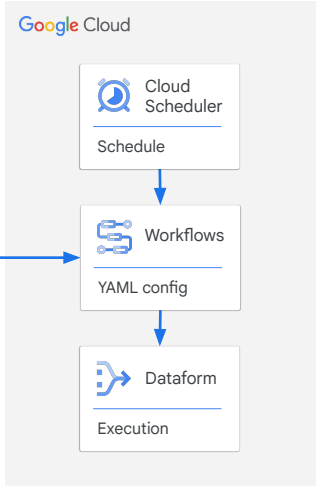
It grants you the flexibility to define both the frequency and precise time of day for job execution.

Triggers can be based on HTTPS calls, App Engine HTTP calls, Pub/Sub messages, or Workflows.

Proprietary + Confidential

Example: Trigger a Dataform SQL workflow

```yaml
- createCompilationResult:
    call: http.post
    args:
        url: ${"https://dataform.googleapis.com/[...]"}
        auth:
            type: OAuth2
        body:
            gitCommitish: <your_branch>
    result: compilationResult
- createWorkflowInvocation:
    call: http.post
    args:
        url: ${"https://dataform.googleapis.com/[...]"}
        auth:
            type: OAuth2
        body:
            compilationResult: ${compilationResult.body.name}
            invocationConfig:
                includedTags:
                - <your_tag>
    result: workflowInvocation
```



Google Cloud

Cloud Scheduler can be used to trigger a Dataform SQL workflow.

In the example code, a scheduled job in Cloud Scheduler initiates the process, defined in a YAML config file.

The workflow involves two main steps: creating a compilation result from your Dataform code and then triggering a workflow invocation using that result, ensuring only specific parts of your Dataform project execute based on included tags.

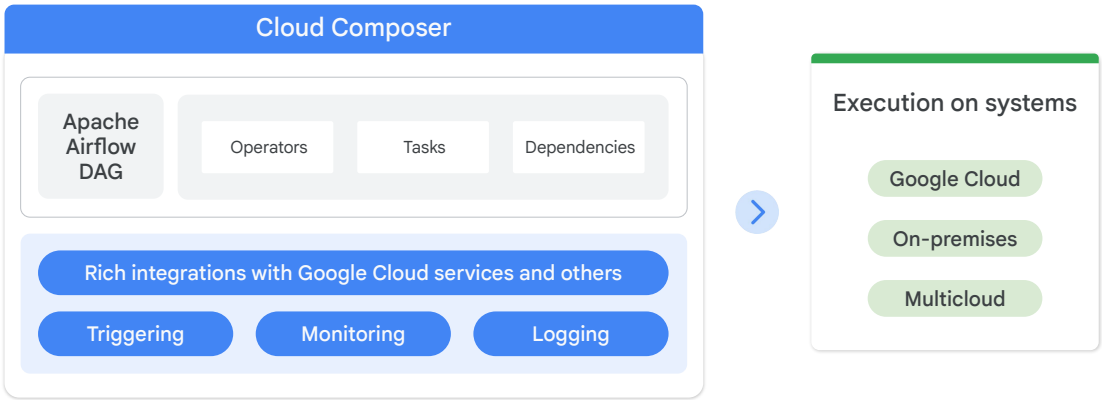
Proprietary + Confidential

Automation techniques

01	Automation patterns and options for pipelines
02	Cloud Scheduler and Workflows
03	Cloud Composer
04	Cloud Run functions
05	Eventarc

Proprietary + Confidential

Cloud Composer orchestrates your pipelines on different systems into workflows



Google Cloud

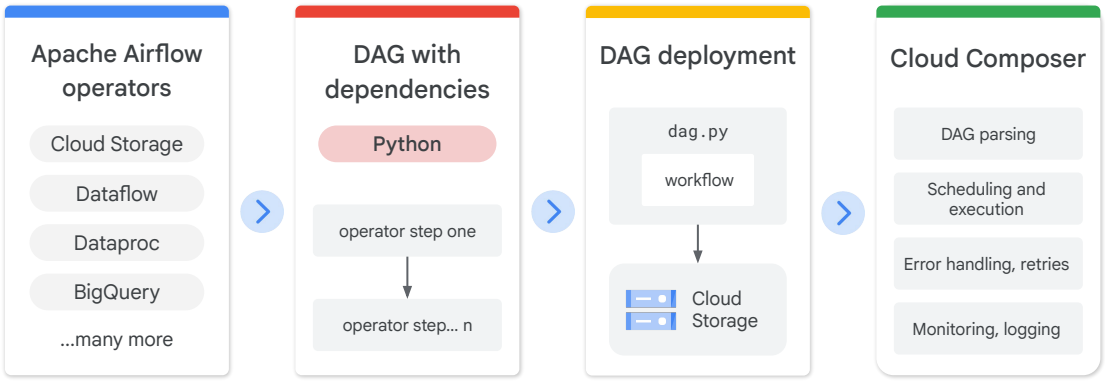
Cloud Composer acts as a central orchestrator, seamlessly integrating your pipelines across diverse systems—whether on Google Cloud, on-premises, or even multicloud environments.

Cloud Composer leverages Apache Airflow, incorporating essential elements like operators, tasks, and dependencies to define and manage your workflows.

Additionally, Cloud Composer offers robust features for triggering, monitoring, and logging, ensuring comprehensive control over your pipeline executions.

Proprietary + Confidential

Develop your DAG in Python using Apache Airflow operators and submit it to Cloud Composer for execution



Google Cloud

Developing and executing workflows using Apache Airflow and Cloud Composer is easily done using Python.

First, you leverage Apache Airflow operators to craft your directed acyclic graph or DAG, defining the tasks and their dependencies.

Next, the DAG is deployed to Cloud Composer, which handles the parsing and scheduling of your workflow.

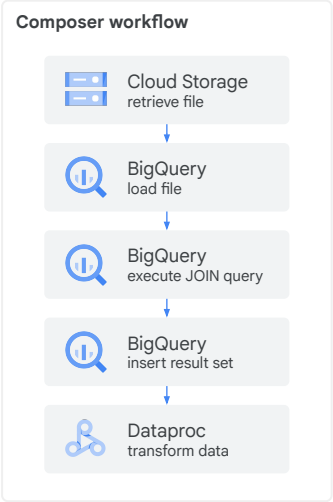
Cloud Composer further manages the execution of your tasks, incorporating features like error handling, retries, and monitoring to ensure smooth operation.

Proprietary + Confidential

Example: Run a data analytics DAG

```
with models.DAG(
   "data_analytics_dag",
   # Define schedule and default args
) as dag:
   create_batch = DataprocCreateBatchOperator(
       # Specify Dataproc settings, e.g. which
       # Python file to execute
   )
   load_external_dataset = GCSToBigQueryOperator(
       # Specify Cloud Storage source file to load
       # and BigQuery table destination
   )
   with TaskGroup("join_bq_datasets") as bq_join_group:
       # Define the SQL query in BigQuery to join
       # the loaded table with another one
           bq_join_holidays_weather_data = BigQueryInsertJobOperator(
               # Execute query and insert result into BigQuery table
           )

       # define the dependencies of the workflow
       load_external_dataset >> bq_join_group >> create_batch
```



Google Cloud

With minimal effort, Cloud Composer can be used to run a data analytics DAG.

In the example code, the workflow retrieves a file from Cloud Storage, loads it into BigQuery, and then performs a JOIN operation with an existing BigQuery table.

The joined results are then inserted into a new BigQuery table.

Finally, Dataproc is used for further data transformation.

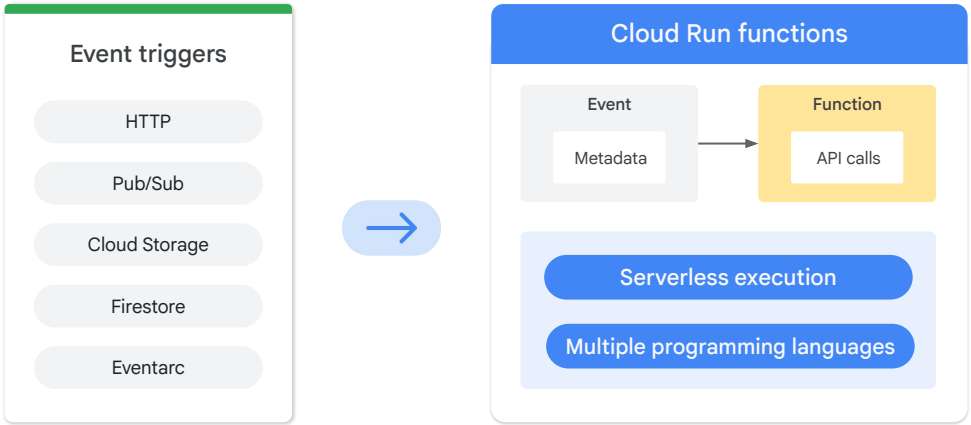
Proprietary + Confidential

Automation techniques

01	Automation patterns and options for pipelines
02	Cloud Scheduler and Workflows
03	Cloud Composer
04	Cloud Run functions
05	Eventarc

Proprietary + Confidential

Use Cloud Run functions to execute code based on Google Cloud events



Google Cloud

Cloud Run functions allow you to execute code in response to various Google Cloud events.

These events can originate from sources like HTTP requests, Pub/Sub messages, Cloud Storage changes, Firestore updates, or custom events through Eventarc.

When triggered, a Cloud Run function provides a serverless execution environment where your code runs, supporting multiple programming languages for flexibility.

Proprietary + Confidential

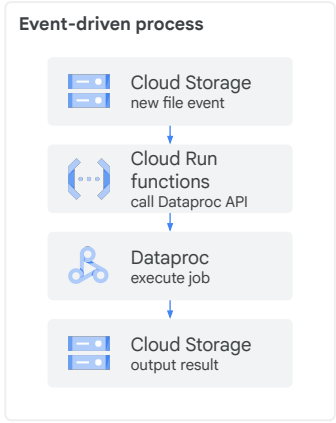
Example: Trigger a Dataproc workflow template after a file upload to Cloud Storage

```
// pre-work: define project ID, workflow template, region
// set up Dataproc API client in specific region
const client = new dataproc.WorkflowTemplateServiceClient({
  apiEndpoint: `${region}-dataproc.googleapis.com`,
});

// retrieve bucket and name of new object on Cloud Storage
const file = data;
const inputBucketUri = `gs://${file.bucket}/${file.name}`;

// construct request to Dataproc API
const request = {
 name: client.projectRegionWorkflowTemplatePath(projectId, region, workflowTemplate),
 parameters: {"INPUT_BUCKET_URI": inputBucketUri}
};

// call API to launch the workflow
client.instantiateWorkflowTemplate(request)
 .then(responses => {console.log("Launched Dataproc Workflow:", responses[1]);
 })
 .catch(err => {console.error(err);
 });
```



Google Cloud

Cloud Run functions easily automate routine tasks on Google Cloud.

In the example code, a Dataproc workflow template is triggered after a file is uploaded to Cloud Storage.

A Cloud Run function is used to capture the Cloud Storage new file event and call the Dataproc API.

The Dataproc API then executes the specified workflow template, using the uploaded file as an input parameter.

The final result of the workflow execution is stored in Cloud Storage.

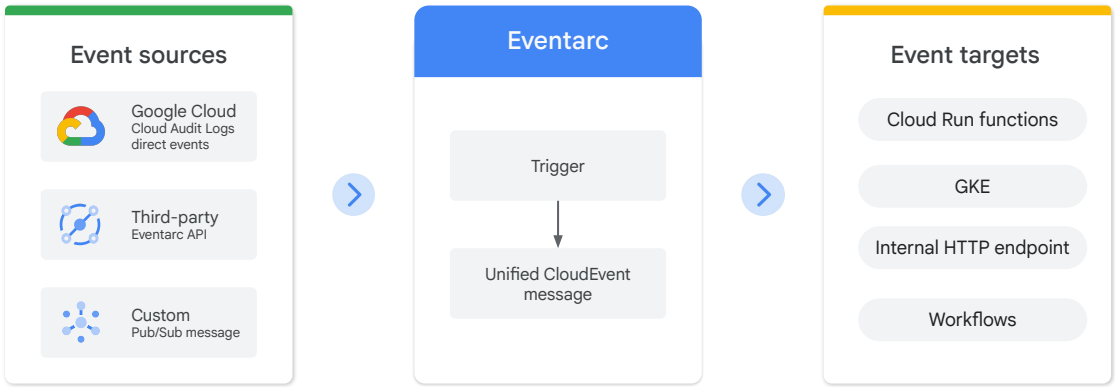
Proprietary + Confidential

Automation techniques

01	Automation patterns and options for pipelines
02	Cloud Scheduler and Workflows
03	Cloud Composer
04	Cloud Run functions
05	Eventarc

Proprietary + Confidential

Build a unified event-driven architecture for loosely coupled services with Eventarc



Google Cloud

Eventarc enables the creation of a unified event-driven architecture for loosely coupled services.

Eventarc connects various event sources, including Google Cloud services, third-party systems, and custom events via Pub/Sub, to a range of event targets like Cloud Run functions, and more.

By using a standardized CloudEvent message format, Eventarc simplifies the integration of diverse systems and facilitates the development of responsive, scalable applications.

Proprietary + Confidential

Example: Respond to `INSERT` events in BigQuery

Cloud Audit Logs event as Eventarc trigger:

Google Cloud

Eventarc enables deep monitoring of logging and other events which occur less frequently on Google Cloud.

In the example code, Eventarc is used to trigger actions in response to data insertion events in BigQuery.

When an insert operation occurs in a BigQuery table, it generates a Cloud Audit Log event.

Eventarc can capture this event and initiate various actions such as rebuilding a dashboard, retraining an ML model, or executing any other custom action based on the specific requirements.

Proprietary + Confidential

Compare automation options

	Cloud Scheduler	Cloud Composer	Cloud Run functions	Eventarc
Trigger type	schedule, manual	schedule, manual	event	event
Serverless	yes	no	yes	yes
Coding effort	low	medium	high	high
Programming languages	YAML (with Workflows)	Python	Python, Java, Go, Node.js, Ruby, PHP, .NET core	any

Google Cloud

In summary, there are various Google Cloud data-related automation options.

Cloud Scheduler and Cloud Composer are suitable for scheduled or manual triggers, while Cloud Run functions and Eventarc are event-driven.

Cloud Scheduler offers low coding effort with YAML, and Cloud Composer requires medium effort with Python.

Cloud Run functions support multiple languages, while Eventarc is language-agnostic.

As a final note, all options except Cloud Composer are serverless.

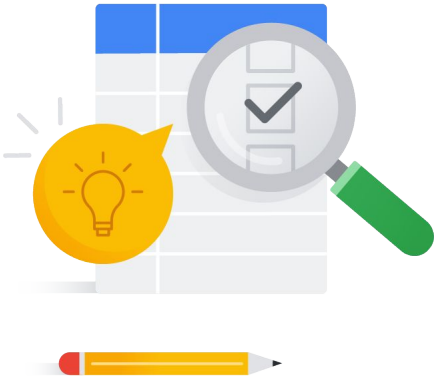
Proprietary + Confidential

Lab: Use Cloud Run Functions to Load BigQuery

45 min

Learning objectives

- Create a Cloud Run function.
- Deploy and test the Cloud Run function.
- View data in BigQuery and review Cloud Run function logs.



Google Cloud

In this lab, you create a Cloud Run function to load BigQuery.

You create a Cloud Run function using the Google Cloud SDK.

You then deploy and test the Cloud Run function.

Finally, you view data in BigQuery and review Cloud Run function logs.